



Andrea Simioni

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Date of birth: 26/05/1994 **Nationality**: Italian

ABOUT ME

I graduated with full marks in computer engineering from the university of Florence. My interests are in machine learning, data science and human-computer interaction. I produce electronic music and I am passionate about photography. Currently i'm working as R&D specialist and software developer in Sintra Digital Business.

WORK EXPERIENCE

Sintra Digital Business

City: Arezzo | **Country**: Italy

[Current] **R&D Specialist and Software Developer**

R&D Specialist

Focus: Researching new technologies, developing software, and creating innovative proof-of-concepts, focusing mostly on data science projects.

Main Projects:

- **Dashboard PoC (Google Data Studio, Power BI, Azure)**: Data aggregation, sales forecasting, and customer segmentation.
- **Automation Platform (Pimcore, Azure AKS)**: Software deployment automation and cloud infrastructure creation.
- **iPaaS (Backend API, .NET, RabbitMQ, MongoDB)**: Data integration for event-driven and scheduled processes.
- **Customer Churn Prediction**: Preprocessing and experimentation with ML/DL models using high unbalanced datasets.
- **Retrieval Augmented Generation (RAG) and Agentic AI**: AI Agents development using private LLM/Microsoft Copilot Studio and vector databases.
- **Corporate Tutor**: Supervised thesis projects for students from the University of Siena and Florence, including sales forecasting, RPA systems for fraud protection and GPT integrations.

Teaching Experience:

Lecturer at the University of Tor Vergata and the University of Pisa on deep learning and practical applications in image classification.

Software Developer

Development and maintenance of software solutions for data integration in the e-commerce and cloud sectors.

Main Projects:

- **Event-Driven Integrations (Azure, Shopify API)**: Managing customers, orders, and products; massive synchronization of prices and inventories between sales channels and external systems.
- **Full-Stack Development (iPaaS, .NET, Vue.js)**: Integrated into the product team since 2024 for full-stack development of the company's iPaaS platform.

Technical Skills:

- **Languages:** C#, Python, PHP, SQL, JavaScript, Bash
- **Databases:** MySQL, MongoDB, Azure Storage
- **Cloud and DevOps:** Azure, Docker, Kubernetes, Helm, Azure DevOps
- **Tools:** Git, Pimcore, Power BI, Google Data Studio, Auth0

EDUCATION AND TRAINING

[19/02/2018 – 29/04/2021]

Master's degree

Università degli studi di Firenze <https://www.unifi.it>

City: Firenze | **Country:** Italy | **Field(s) of study:** Computer engineering | **Final grade:** 110/110 | **Level in EQF:** EQF level 7 | **Number of credits:** 123 | **Thesis:** SAR image despeckling by means of deep learning methods

Key Projects and Analyses:

- **Convolutional Neural Network Development:** Created a CNN for denoising optical images and those from remote sensing systems like Synthetic Aperture Radar (SAR).
- **Variational Autoencoder for Music Generation:** Developed a model for generating multitrack polyphonic music in MIDI format.
- **Adversarial Attack Analysis:** Studied adversarial attacks on classifiers, implementing defense mechanisms using JPEG compression and Generative Adversarial Networks (GANs).
- **Object Detection with Deep Learning:** Applied deep learning methods for real-time object detection.
- **Object Trajectory Estimation:** Used Mask-RCNN with the KITTI dataset to estimate object trajectories in automotive contexts.
- **Game Development (Unity 3D):** Designed VR applications, including environment creation and AI development for NPCs and game mechanics.
- **Human-Machine Interaction:** Developed a Leap Gestural Car, integrating Raspberry Pi with gesture control via Leap Motion Controller.

Technical Skills:

- **Mathematics and Statistics:** Strong foundation in mathematical principles.
- **Programming Languages:** Python, Java, and C++ (basics).
- **Libraries and Frameworks:** Experienced with TensorFlow/Keras, PyTorch, Scikit-learn, and OpenCV.

[08/02/2022 – 20/05/2022]

Executive Master, Data Science, Big Data & Artificial Intelligence per la Finanza

24ORE Business School <https://www.24orebs.com>

Field(s) of study: Data Science | **Thesis:** Stock price prediction algorithm in Python from financial historical series using ARIMA models and dense neural networks

main topics addressed:

- main machine learning algorithms
- Deep learning algorithms such as dense, convolutional neural networks and autoencoders
- Real world applications of machine learning and deep learning with focus on financial and big data sector

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING B2 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

AI, Machine learning and deep learning

Deep learning in image processing | Forecasting | Autoencoders | Clustering | YOLO
e YOLO Tiny | Convolutional Neural Networks (CNN o ConvNet) | Basic Data Science
Data Analytics | Dense Neural Networks | Object Detection | regression | Generativ
e Adversarial Networks | Computer Vision | Image Classification

Programming languages and libraries

Python | Bash | TensorFlow | React.js (Basics) | C# | Pandas | Node.js | PHP | J
avascript | C++ (Basics) | Keras | OpenCV | CSS (Basics) | Typescript | Matplotlib
| Numpy | Express.js | Scikit-Learn | Nest.js | ASP.NET | Solidity (Basics) | Java
(Basics) | SQL | HTML

DevOps

Git | Docker | Kubernetes

Database Systems

MongoDB | MySQL | Firebase

Cloud

Azure Data Factory | Azure Functions | Azure Storage | Azure Service Bus | Azure
Container Registry | Azure Kubernetes Services | Azure Machine Learning

Platforms and other tools

Microsoft Power Bi | Pimcore | Google Data Studio | Visual Studio, Android studio,
Unity | Auth0

HOBBIES AND INTERESTS

Music and photography

I produce electronic music mainly related to the techno and ambient genre. I am the co-
owner of Blame Rec, a record label whose aim is to promote the total creative expression
of new artists.

Also, I am very passionate of landscape photography.

Sports and physical activity

I've been going to the gym for 8 years and I've also been doing CrossFit regularly for 3
years. In 2009 I obtained the PADI Advanced Open Water Diver and Enriched Air Diver
certificates.